

BENJAMIN RIOT-BRETÊCHER

Last updated: March 6, 2026

+1 438-505-5702 ◊ Burnside Hall, Room 949 ◊ 805 Sherbrooke St W, Montréal, QC H3A 0B9

benjamin.riotbretacher@mail.mcgill.ca ◊ www.linkedin.com/in/benjamin-rb/ ◊ <https://github.com/binmenja>

EDUCATION

Ph.D - Atmospheric and Oceanic Sciences, McGill University 2022 -

Supervisor: [Prof. Yi Huang](#)

Relevant Coursework: Atmospheric Dynamics, Atmospheric Radiation, Numerical Methods, Waves & Stability

ACHIEVEMENTS: Comprehensive Examination (February 2024); Proposal Examination (August 2025)

B.S - Atmospheric and Oceanic Sciences, Université du Québec à Montréal 2019 - 2022

PROFESSIONAL EXPERIENCE

Teaching Assistant 2023 -

McGill University

- [Introduction to Atmospheric Sciences - ATOC181](#) (Fall 2023, Fall 2025)
- [Science of Storms - ATOC184](#) (Winter 2024, Winter 2026)

Visitor Student Oct 2024 - Jan 2025

[School of Atmospheric Sciences, Nanjing University](#)

Treasurer Aug 2023 - Aug 2024

Council of Atmospheric and Oceanic Sciences (CAOS)

Field Campaign Involvement 2022 - Present

McGill University

- [ECALOT Field Campaign](#) (Fall 2024)
- [WHAFFERS](#) (Lead student investigator; Winter 2025)
- [GHG Montreal Project](#) (Ongoing)
- [PONEX](#) (Lead student investigator; Winter 2026)

Research Assistant May 2022 - Aug 2022

McGill University

Supervisor: [Prof. Yi Huang](#)

Research Assistant June 2021 - Aug 2021

Université du Québec à Montréal

Supervisor: [Prof. Jean-Pierre Blanchet](#)

PUBLICATIONS

- **B. Riot-Bretêcher**, Y. Huang, L. Liu, J. Gero, [Strengthening Atmospheric Greenhouse Effect at the Arctic North Slope of Alaska Site Evidenced by Long-Term Records of the Downwelling Longwave Radiance Spectrum](#), Journal of Geophysical Research: Atmospheres, 2025.
- Quinn Dyer-Hawes, Djordje Romanic, Yi Huang, John R. Gyakum, Peter M. J. Douglas, Lei Liu, **B. Riot-Bretêcher**, Regina Gonzalez Moguel, Jacob Asomaning, John Stix, Gustavo Villarruel, Margaret Kalacska, Oliver Lucanus, Juan Pablo Arroyo-Mora, Keyvan Ranjbar, Leonid Nichman, Felix Vogel, *Analysis of urban greenhouse gas dispersion for downtown Montréal using computational fluid dynamics and field measurements*, Atmospheric Pollution Research (submitted).
- Y. Wang, G. X. He, **B. Riot-Bretêcher**, M. Pelletier, Y. Huang, S. K. Loeb, *Integrating Meteorological Data into Linear Mixed Models for Sunlight Inactivation of Indicator Organisms in Natural Lake Water* (in prep)
- B. Pery, H. Brindley, J. Murray, R. Hogan, S. Mosselmans, S. Panditharatne, L. Warwick, T. L'Ecuyer, **B. Riot-Bretêcher**, Y. Huang, *Towards ground-to-space radiative closure in the mid- to far-infrared* (in prep)
- Z. Qu, C. Nguyen, P. Borque, A. Korolev, H. Barker, K. Ranjbar, L. Liu, Z. Mariani, M. Kacimi, M. Cholette, I. Heckman, K. Bala, N. Bliankinshtein, L. Nichman, **B. Riot-Bretêcher**, Y. Huang, J. Cole, R. Peroni, S. Holden, R. Reed, R. Crawford, D. Michelson, E. Bigras, M. Wheeler, J. Gyakum, J. Milbrandt, J. Thériault, K. Walker, J.-P. Blanchet, Y. Blanchard, *EarthCARE Commissioning Calibration/Validation Campaign in Ottawa (ECALOT)* (in prep)

CONFERENCES AND TALKS

- [Joint FORUM-HAWC Workshop - Poster](#) - Montreal, QC, Canada - October 2025

- [Gordon Research Conference on Radiation and Climate](#) - **Invited Speaker** - Lewiston, ME, USA - July 2025
- [Gordon Research Seminar on Radiation and Climate](#) - **Speaker** - Lewiston, ME, USA - July 2025
- [2nd ESA-JAXA EarthCARE In-Orbit Validation Workshop](#) - **Co-author, Presented by L. Liu** - Frascati, Italy - March 2025 Validation of EarthCARE cloud retrievals using surface spectral infrared radiances from the ECALOT campaign. (Liu, L., **Riot-Bret cher, B.** et al.)
- [Nanjing University - Atmospheric Sciences Seminar](#) - **Speaker** - Nanjing, China - December 2024
- [43rd Canadian Symposium on Remote Sensing](#) - **Attendee** - Qu bec City, QC, Canada - July 2022
- [Canadian Meteorological and Oceanographic Society](#) - **Attendee** - Remote - June 2021

AWARDS AND SCHOLARSHIP

- McGill Graduate Excellence Fellowship (2022 - 2028)
- Graduate Mobility Award - McGill University (Oct 2024 - Jan 2025) - \$ 3,800
- Canadian Meteorological and Oceanographic Society Undergraduate Scholarship (April 2021) - \$ 1,000

SKILLS

Technical Skills	MATLAB, Fortran, Python, Java, R, Bash, Linux, Radiative Transfer Model, Weather Balloon Launches, AERI Maintenance, Optimal Estimation Method
Mathematical Skills	Algebra, Calculus, Probability, Statistics
Certifications	• Data Security - Compute Ontario (June 2025) • Introduction to Advanced Research Computing - Compute Ontario (June 2025) • Introduction to Version Control Using Git - Compute Ontario (June 2025) • Scaling Up HPC Workflows - Compute Ontario (June 2025) • Career Essentials in Software Development by Microsoft and LinkedIn (Certificate ID) • Programming Foundation: Beyond the Fundamentals by LinkedIn (Certificate ID)